

Mark schemes

Q1.

(a) (i) $\overrightarrow{AB} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \quad \overrightarrow{AD} = \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$
either $\overrightarrow{AB}$ or $\overrightarrow{AD}$

B1

$$\begin{vmatrix} \mathbf{i} & 2 & 4 \\ \mathbf{j} & -1 & 3 \\ \mathbf{k} & 3 & -1 \end{vmatrix} = \begin{pmatrix} -8 \\ 14 \\ 10 \end{pmatrix}$$

one component of $\overrightarrow{AB} \times \overrightarrow{AD}$ correct

M1

all correct

A1cao

3

(ii) Area $ABCD = |\overrightarrow{AB} \times \overrightarrow{AD}| = \sqrt{8^2 + 14^2 + 10^2}$
FT their $|\overrightarrow{AB} \times \overrightarrow{AD}|$

M1

$$= \sqrt{64 + 196 + 100}$$

$$= \sqrt{360}$$

$$= \sqrt{36} \sqrt{10}$$

$$= 6\sqrt{10}$$

$$\text{or } p = 6$$

A1cso

2

(b) $\overrightarrow{AB} \times \overrightarrow{AD}$ is perpendicular to plane $ABCD$
used for $\mathbf{v}$ in vector line equation

M1

Hence direction ratios of line = $-8 : 14 : 10 = -4 : 7 : 5$

$$\text{or any multiple of } \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$$

A1

M is mid-point of either diagonal = $\left(\frac{1+7}{2}, \frac{0+2}{2}, \frac{2+4}{2}\right)$
mid-point calculation

B1

$$= (4, 1, 3)$$

Hence line is $\left(\mathbf{r} - \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}\right) \times \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

All correct or equivalent multiple of $\begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$

A1cso

4

- (c) Perpendicular vector to $\Pi = \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$
 $\Rightarrow \Pi$ has equation $-4x + 7y + 5z = c$
ft their perpendicular vector

M1

Through $(6, 5, 17) \Rightarrow c = -4(6) + 7(5) + 5(17)$
using $(6, 5, 17)$

$$= -24 + 35 + 85 = 96$$

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A1

Equation is $-4x + 7y + 5z = 96$
ACF

$x = 4 - 4t; y = 1 + 7t; z = 3 + 5t$
parametric form of line

B1F

Line meets plane when
 $-4(4 - 4t) + 7(1 + 7t) + 5(3 + 5t) = 96$
 $-16 + 16t + 7 + 49t + 15 + 25t = 96$
 $90t = 90$

$$t = 1$$

substitution of parametric form and attempt to solve for t

M1

⇒ point of intersection = (0, 8, 8)
correct point of intersection

A1cao

6

Alternative 1:

$$\mathbf{r} = \begin{bmatrix} 6 \\ 5 \\ 17 \end{bmatrix} + s \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$$

$\mathbf{r} = \mathbf{a} + s\mathbf{d}_1 + t\mathbf{d}_2$ fully correct

(M1)

$$\begin{vmatrix} i & 2+2s+4t & -4 \\ j & 4-s+3t & 7 \\ k & 14+3s-t & 5 \end{vmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Substitute in $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = 0$

(m1)

Use of $\mathbf{r} - \mathbf{a}$ in parametric form – simplified

(B1F)

$$-26s + 22t = 78$$

$$22s + 16t = -66$$

$$10s + 40t = -30$$

Three correct equations obtained from vector product-terms collected

(A1)

$$s = -3, t = 0$$

Correctly solving equations to get both parameters

(M1)

$$\begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix}$$

Correct point of intersection

(A1)

(6)

Alternative 2:

$$\mathbf{r} = \begin{bmatrix} 6 \\ 5 \\ 17 \end{bmatrix} + s \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$$

$\mathbf{r} = \mathbf{a} + s\mathbf{d}_1 + t\mathbf{d}_2$ fully correct

(M1)

$$\mathbf{r} = \begin{pmatrix} 4-4p \\ 1+7p \\ 3+5p \end{pmatrix}$$

Parametric form of line

(B1F)

$$2s + 4t + 4p = -2$$

$$-s + 3t - 7p = -4$$

$$3s - t - 5p = -14$$

Equating components, simplifying and attempting to solve-must at least reduce to 2 equations in two unknowns

(m1)

$$p = 2$$

Solving equations-one parameter correct

(M1)

$$t = 0 \text{ and } s = -3$$

All values correct

(A1)

$$\begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix}$$

Correct point of intersection

(A1)

(6)

[15]

Q2.

$$(a) \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix} = 12 + 15 - 16 = 11 \text{ shown}$$

B1

1

(b) eqn., or use, of line incorporating

$$\text{p.v. } \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \text{ and d.v. } \begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix}$$

M1

$$\frac{x-1}{12} = \frac{y-1}{15} = \frac{z+1}{16}$$

A1

2

(c) Use of $\mathbf{r} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix}$ with $\lambda = \pm 2$

M1

$(25, 31, 31)$ and $(-23, -29, -33)$

A1 A1

3

(d) **Method 1**

$PQ = 5$

B1

Area $\triangle PMN = \frac{1}{2} \times PQ \times MN = 250$

(Since $Q = \text{midpt. } MN$)

M1 A1

3

Method 2

$\overrightarrow{PM} = \begin{bmatrix} 20 \\ 30 \\ 35 \end{bmatrix}, \overrightarrow{PN} = \begin{bmatrix} -28 \\ -30 \\ -29 \end{bmatrix}$

Attempted

(M1)

$\overrightarrow{PM} \times \overrightarrow{PN} = (\pm) 20 \begin{bmatrix} 9 \\ -20 \\ 12 \end{bmatrix}$ attempted

$||gm. \text{ or } \Delta$

(M1)

within an area formula

Area $\triangle PMN = 250$

CAO

(A1)

[9]

Q3.

For attempt at 2 of $(\pm) \overrightarrow{PQ}, \overrightarrow{PR}, \overrightarrow{QR}$

M1

$$\begin{aligned} \text{Area } \Delta PQR &= \frac{1}{2} |\overrightarrow{QP} \times \overrightarrow{QR}| \text{ e.g.} \\ &= \frac{1}{2} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 2 & -3 \\ 2 & 0 & -2 \end{vmatrix} = \frac{1}{2} |\pm (4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k})| \end{aligned}$$

*Formula used with attempt at a vector product
of any 2 of the above (ignore missing $\frac{1}{2}$ for now)*

M1

$$= \frac{1}{2} \sqrt{4^2 + 2^2 + 4^2}$$

Method for finding magnitude of their relevant vector

M1

$$= 3$$

CSO

A1

[4]

Q4.

$$(a) \quad \mathbf{p} \times \mathbf{q} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 2 \\ -3 & 4 & 40 \end{vmatrix} = \begin{bmatrix} 4 \\ -32 \\ 7 \end{bmatrix}$$

M1 A1

2

$$(b) \quad A = \frac{1}{2} |\mathbf{p} \times \mathbf{q}|$$

M1

$$= \frac{1}{2} \sqrt{4^2 + 32^2 + 7^2}$$

For attempt at $|\mathbf{p} \times \mathbf{q}|$

B1

$$= \frac{33}{2}$$

ft

A1F

3

[5]

Q5.

(a) $\mathbf{b} - \mathbf{a} = 4\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$ $\mathbf{c} - \mathbf{a} = \mathbf{i} - 10\mathbf{j} + 2\mathbf{k}$

Either correct

B1

$$(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -6 & 2 \\ 1 & -10 & 2 \end{vmatrix}$$

Genuine attempt using their two vectors

M1

$$= 8\mathbf{i} - 6\mathbf{j} - 34\mathbf{k}$$

CSO

A1

3

(b) Area $\triangle ABC = \frac{1}{2} |\text{this vector}|$

Must be "Hence" method

M1

$$= \frac{1}{2} \times 2\sqrt{4^2 + 3^2 + 17^2}$$

Correct modulus attempt

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M1

$$\sqrt{314} \text{ or } 17.7(2)$$

ft (b)(i) only

A1✓

3

[6]

Examiner reports

Q1.

Generally there was good response, with all students able to score marks in the earlier part of the question. The vector product evaluation through a determinant is well known; nevertheless, a handful of students made sign errors. Students also know how to apply the determinant to finding the area of a parallelogram. Many students did not appreciate the link between the parts of this question. Thus the determinant evaluated in part (a) was re-evaluated in part (b) albeit with the diagonals of the parallelogram instead. With regard to vector notation, students should realise that they must make key vectors clear, eg the

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

zero vector written as $\mathbf{0}$ or $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Many students did not find the coordinates of $\mathbf{M}$ by using the mid-point result from Core 1 but instead chose to find the intersection point of two lines. Part (c) proved most challenging. Those who used the scalar product or Cartesian form of the plane achieved the greatest success in finding the point of intersection of the line and plane.

Q2.

Around a quarter of all candidates made attempts only at part (a), for just the one mark, and around half the candidates failed to proceed beyond part (b). Even amongst these, there was a large number of candidates who apparently didn't know the difference between vector and cartesian equations, with many others not happy to work with the given normal vector.

Of that half of the candidature that did proceed beyond part (b), most of them decided that "a distance of 50 units" meant that λ was ± 50 , rather than ± 2 .

Even in the final part of the question, despite being given the coordinates of P and Q and the information that the line PQ was the height of a right-angled triangle with base 100, only about five candidates (of 336) appreciated that they could work out the area of the triangle just as it was. Even more disappointing than this was that so few could employ alternative vector methods to get the correct answer, mostly due to not having had much success with parts (a) to (c).

Q3.

This was generally done very well. The only common mistake arose when candidates quoted the area formula for a triangle as $\frac{1}{2} |\mathbf{a} \times \mathbf{b}|$ but then thought that $\mathbf{a}$ and $\mathbf{b}$ were $\mathbf{p}$ and $\mathbf{q}$ (for instance) rather than $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ (say).

Q4.

This question was handled very well indeed, apart from minor slips in the arithmetic.

Q5.

This was one of the most successfully attempted questions on the paper for most candidates, although marks were often lost through a lack of care with signs somewhere

along the line. Many forgot the factor of $\frac{1}{2}$ in part (b), but otherwise this was a high-scoring question for the majority of candidates.



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